

Analysis of Various Traffic Sign Detectors Based on Deep Convolution Network

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Abstract—A lot of progress has been made on the object detection field, and it is difficult for researchers to decide which architecture is suitable to their specific task. The work in this paper aims to compare and analyze various detection models and then obtain an appropriate model for the traffic sign detection task. Two main components of a detection model are meta-architecture and feature extractor, and we will explain these two parts on the paper. Experiments show that the behavior of the detection model is strongly affected by the choice of architecture and feature extractor. Also, Region-based Fully Convolutional Networks (R-FCN) architecture can achieve better performance than Faster Regions with Convolutional Neural Network (Faster R-CNN) and Single Shot Detection Network (SSD) in traffic sign detection task.

Index Terms—detection architecture, traffic sign detection, tensorflow

I. INTRODUCTION

With the popularity of driver-assistance system embedded in vehicles, traffic sign recognition (TSR) has been receiving more and more attention in recent years. TSR is a technology which can provide semantic information about different traffic signs on the road, such as "speed limit" or "turn ahead" or "stop". TSR technology is used in a wide range of application from driver-assistance systems [5] to a fully autonomous vehicle [6].

Generally, TSR consist of two parts: traffic sign detection and traffic sign classification. The traffic sign classification technology has yielded superhuman performance [7]. Detection is a prerequisite of classification. Also compared to traffic sign classification, traffic sign detection that requires more complex methods to solve is a more complicated task and still has much room for improvement. The Complexity of detection arises because detection requires two steps to obtain accurate localization of objects. Firstly, candidates that only provide rough localization of objects must be obtained from numerous proposals. Secondly, coarse localization must be refined to achieve precise localization. [4].

Recently, a lot of progress has been made on object detection [3] and classification [1] [2] due to deep ConvNets [1], and many successful detection network architectures have been proposed, such as R-CNN [9] [4] [10], R-FCN [11], SSD [20], and YOLO [13] [14] [15]. Because a wide variety of architectures have been proposed and has its unique

speed to accuracy ratio [16], it is difficult for a researcher to decide what architecture and configuration are best suited to specific task, such as different feature extractors (e.g., Inception [17], Residual Network [18]), different detection architectures (e.g., Faster R-CNN, SSD).

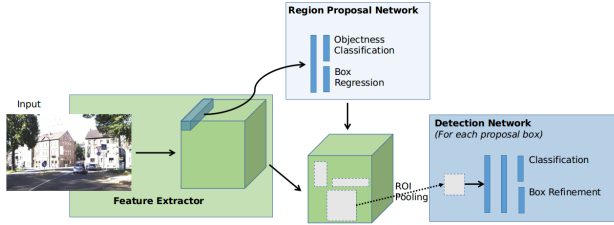
Each specific task has characteristics that can be used to select specific detector to improve performance. Also, the no free lunch theorem implies that we should design our machine learning algorithm to perform well on a specific task. It is easy for human and detector to notice the location of traffic signs because traffic signs are designed in specific shapes and high saturated colors, also the symbols and text of traffic sign are distinctive easily to its surrounding background. In the meanwhile, because the size of the traffic sign is relatively small in an image, the detection architectures have inferior performance [16] on small objects.

The critical contributions of this paper are that 1) the experiment shows a comparison of 5 state-of-the-art detection methods, and the paper can be served as a guide for selecting a detection architecture for traffic sign detection task 2) the paper details the benefits and drawbacks of each meta-architecture, which can help the readers selecting a proper detection architecture for their specific tasks and 3) the paper demonstrates the R-FCN can achieve better performance in target occluded case.

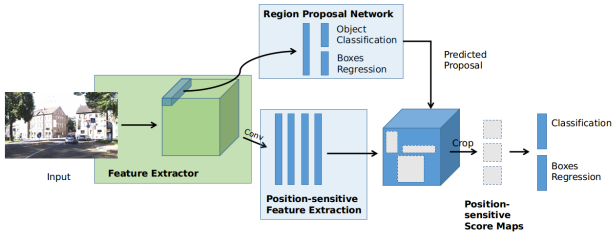
In this paper, we investigate various traffic sign detector using Tensorflow object detection API (Application programming interface) [16], which enable apples-to-apples comparisons more straightforward. The rest of this paper is organized as follows. Section II presents three meta-architectures. Experimental setup and analysis are further presented in Section III. Conclusions are finally drawn in Section IV.

II. DETECTION META-ARCHITECTURES

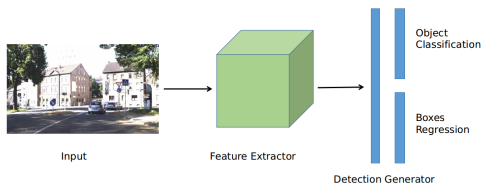
Compared to full image classification, detection models tend to be more complex. There are mainly two types of the popular object detection algorithms, two-stage and one-stage algorithm. Figure 1 shows three types of meta-architecture, (a) Faster-RCNN, and (b) R-FCN are corresponding to the two-stage algorithm, (c) SSD belongs to the one-stage algorithm. The two-stage algorithm includes that (i) use a Region Proposal Network (RPN) to generate some coarse boxes and



(a) Faster R-CNN.



(b) R-FCN.



(c) SSD.

Fig. 1. The diagrams of Meta-architectures in this paper.

(ii) use the following detection network to refine these coarse boxes. The one-stage algorithm learns the class probabilities and bounding box coordinates of the boxes on several layers of the network to produce the predicted boxes, it followed by a non-maximum suppression step to remove the redundant boxes. The two-stage algorithm produces the predicted boxes first and then it uses the following detection network to refine the boxes. So compared with the one-stage algorithm, the two-stage algorithm tends to achieve higher accuracy but lower computational efficiency.

Tensorflow Detection API [16] is a platform that can help researchers compare each detection model apples-to-apples, and then researchers can choose a suitable detection model for specific task. In this paper, we used Tensorflow Detection

API to manage the experiments, so we focus on three modern architectures that API provides: Faster-RCNN [10], R-FCN [11] and SSD [20], the feature extractor we used are Inception_v2, Resnet, and the combination of Inception_v2 and Resnet. Any feature extractor can be used with each meta-architecture.

A. Faster R-CNN (Faster Regions with Convolutional Neural Network)

Faster R-CNN is the updated version of R-CNN and Fast-RCNN, R-CNN was proposed in 2014, and it is an early application of CNN (Convolution Neural Network) for object detection. R-CNN has led to many follow-up works, such as Fast R-CNN and Faster R-CNN. In R-CNN, detection is happened in two stages. The First step, Selective Search [19] was used to obtain the region proposals; The second step, features of each proposal has been extracted by passing the proposals to a ConvNet, and then that a Support Vector Machine (SVM) is attached on the final layer of detection will classify the proposal.

R-CNN achieves dramatic detection accuracy by using a deep ConvNet to detect the target object, but it takes a massive amount of time to train the network. R-CNN is slow because it inputs each object proposal to a ConvNet, without sharing computation. Instead of feeding each region proposals to feature extractor, Fast R-CNN [4] which was proposed in 2015 improves training and testing speed by inputting the whole image to the feature extractor. Fast R-CNN joint training, classification and bounding box regression steps in a single neural network model for reducing processing time. Compared to R-CNN, Fast R-CNN is $9\times$ faster at training-time and $214\times$ faster at test-time [4], it still adopt selective search to extract region proposal which still consume a lot of time.

Faster R-CNN was proposed in 2015, and it further reduces the processing time of the object detection task. The architecture of Faster R-CNN is shown in Figure 1(a). Instead of using Selective Search, Faster R-CNN presents an RPN (Region Proposal Network) to generate the coarse proposals. Compared to selective search that is used in R-CNN and Fast R-CNN, RPN can share full-image convolution feature with the following detection network, and it is trained end-to-end to output high-quality proposals. Faster R-CNN achieve almost $10\times$ faster than Fast R-CNN at entire object detection process [10]. Many follow-up works of FasterR-CNN, including SSD and R-FCN have been successively proposed.

B. R-FCN (Region-based Fully Convolutional Networks)

R-FCN [11] has been proposed in 2016, the architecture of R-FCN is shown in Figure 1(b). There are 2 significant contributions of R-FCN. Firstly, instead of standard ROI pooling operation that used in Faster R-CNN, the author of R-FCN proposed a position-sensitive cropping mechanism. Secondly, compared to Faster R-CNN, all of the convolution weights are shared in R-FCN [11], all learnable layers (e.g. convolution layer) are shared on the entire image, and as



Fig. 2. Sample traffic sign images from Germany Traffic Sign Detection Benchmark [8].

shown in figure 1, all of the convolution layer have been done before cropping. Therefore R-FCN is faster than Fast R-CNN or Faster R-CNN. Also, position-sensitive score maps make object detection more precise, because these maps specialized in detecting separate parts of an object.

C. SSD (Single Shot MultiBox Detector)

SSD [20] has been proposed in 2015 and published in 2016, the architecture of SSD is shown in Figure 1(c). Unlike Faster R-CNN and R-FCN which require two steps to detect objects, SSD uses a single feed-forward convolutional network to predict classes and obtain boxes regression value. Therefore faster R-CNN and R-FCN tend to obtain higher detection accuracy but slower than SSD. SSD performs worse than Faster R-CNN when detecting small-scale objects [16] because small objects only exist in the first few layers and it may disappear in the last few layers. Also, the first few layers of the network contain low-level features that are less informative for classification. The experiment is shown in section III, SSD shows the worst performance among these three architectures (SSD, Fast R-CNN, and R-FCN) in traffic sign detection task because most of the scale of the traffic signs are relatively small on the entire input image.

III. EXPERIMENT AND ANALYSES

A. Experiment Setup

The central challenge in traffic sign detection task is that some of the input images are captured under the challenging condition. As shown in figure 2, there are 8 sample input images from GTSDDB dataset. The images from GTSDDB [8] dataset captured under varying conditions, such as poor/strong lighting, different weather condition, relative small-scale sign, and complicated background. The images of traffic sign from Open Image Dataset V4 [21] are clear and relatively large-scale in an image. Compare to GTSDDB, the Open Image Dataset V4 is rather different from real road environment. In this paper, we use GTSDDB dataset to evaluate the model's performance because we want to simulate traffic sign recognition system (TSR) of the autonomous vehicle

because using similar dataset can keep the generalization error low in practice as well.

There are 900 images in GTSDDB dataset. We split 900 images into training and testing dataset in ratio 8:2, we use 720 images to train and 280 images to test the model. GTSDDB dataset contains 1232 labelled traffic signs from the various scales and viewing angles. Figure 2 shows 8 sample traffic sign images which meet challenging situations such as, the shade of trees on the road, relatively small-scale signs, bad weather, the complexity of the road alignment, strong illumination, and insufficient light. Some images do not contain any traffic sign. We show this case in the last two images of the second row of figure 2.

A pre-trained network is re-trained using GTSDDB dataset and no data augmentation algorithm has been used. The RM-SProp optimizer is used on training step with 0.001 learning rate and 240,000 training steps for 5 models, which are shown in figure 1. The performance is analyzed by detecting all the 1232 traffic signs in GTSDDB dataset. Also, in this paper, we use the Tensorflow Object Detection API, which is an open source framework built on top of Tensorflow, all of the detection models that are mentioned in the paper were pre-trained on the COCO dataset [22]. We apply different convolutional feature extractor for three types of the meta-architectures (Faster R-CNN, R-FCN, and SSD) which are introduced in Section II. With different meta-architecture and feature extractor, we evaluate the performance of five types of detection models in total, the experimental results are shown in Table I.

B. Experimental Result

Table I shows the precisions corresponding to each model, and the experiment prove that the architecture of R-FCN can obtain highest accuracy among these five models in traffic sign detection task. Figure 3 is some example images of experimental result of different (row) testing images and (column) models. The first column shows the result used *ssd_inception_v2* model; the second column is the result of *faster_rcnn_resnet101* model; the third column is the result of *faster_rcnn_resnet_50* model, and the fourth and



Fig. 3. Example images of experimental results for different (row) testing images and (column) models.

TABLE I
CONVOLUTIONAL DETECTION MODEL AND TEST ACCURACY

Model Name	Meta-architecture	Featruer Extractor	Dataset	Test Accuracy
ssd_inception_v2	SSD	inception_v2	GTSDDB	0.7937
faster_rcnn_inception_v2	Faster_RCNN	resnet101+inception_v2	GTSDDB	0.9321
faster_rcnn_resnet50	Faster_RCNN	resnet50	GTSDDB	0.9297
faster_rcnn_resnet101	Faster_RCNN	resnet101	GTSDDB	0.9197
rfcn_resnet101	R-FCN	resnet101	GTSDDB	0.9396

TABLE II
PERFORMANCE OF FAST-RCNN AND R-FCN UNDER DIFFERENT GROUND TRUTH QUALITY

Model Name	Meta-Architectrue	Ground Truth Quality	Dataset	Test Accuracy	
				set1	set2
fast_rcnn_resnet101_v	Fast_RCNN	Visible	STSD	0.9082	0.8947
rfcn_resnet101_v	R_FCEN	Visible	STSD	0.9038	0.8917
fast_rcnn_resnet101_vo	Fast_RCNN	Visible+Occluded	STSD	0.8516	0.8585
rfcn_resnet101_vo	R_FCEN	Visible+Occluded	STSD	0.8790	0.8679

fifth columns are the results of *faster_rcnn_inception_v2* and *rfcn_resnet101*, respectively. The green blocks in images are the detection results, and the value on the top of the green block is the confidence value of detection. There are two speed limit traffic signs in sample 1 image, both of them are detected successfully by every model. There are some previous works prove that detection model can perform better on the mandatory category than danger and prohibitory categories [23] [24]. The location of green blocks which were obtained by *ssd_inception_v2* and *faster_rcnn_inception_v2* are not as accurate as the one by *rfcn_resnet101*.

In sample 2, *faster_rcnn_resnet50* obtains a false-positive region. *ssd_inception_v2* performs worse than when the target shape is diamond (sample 3 and sample 4) and the scale of target is relatively small (sample 4). Also

faster_rcnn_resnet50 and *rfcn_resnet101* cause false-positive detection result which mistake a light spot for traffic sign. The *rfcn_resnet101* causes false-positive detection result because the model may misread a sexangular light spot as a traffic sign. There is a type of sexangular traffic that is called "STOP Sign" in GTSDDB, it may be the reason of this fault. The validity of this theory requires further experiments and more training and testing dataset. Because we used 720 images to train each model and we have 43 classes traffic sign, it means that we only have about 20 images for each class, it is not enough to train the neural network to extract appropriate features of "STOP Sign". The benefit and drawback of each traffic sign detector vary dramatically in different classes, training the model to recognize the class of traffic sign can help us acquire a more particular knowledge of each detector.

We will train and test each model to detect and recognize each class of traffic sign in further research. The reason is that most of the shape of traffic signs are a circle rather than the polygon in real-road as well as GTSDb, also, GTSDb has a limited number of images for training and testing, which makes the accuracy easy to saturate.

As shown in Table I, we see that *rfcn_resnet101* achieved the highest test accuracy (0.9395) compared with other models. As shown in the second and third rows of figure 3 and Table I, SSD model obtain the lowest detection accuracy, and SSD cannot produce the precise output of the traffic sign region. So that SSD model do not perform well on traffic sign compare with Faster R-CNN and R-FCN model. The previous research [16] proves that SSD models do well on large objects but no small, and our experiments also provide hard evidence for the assertion. Compared with different architectures, R-FCN can achieve better result than Faster R-CNN and SSD. Our experiment also shows that the choice of feature extractor is crucial as meta-architecture for model performance. The *faster_rcnn_inception_v2* can achieve higher accuracy. *faster_rcnn_inception_v2* denote as using the combination of resnet101 and inception_v2 to extract feature.

Considering the architecture of R-FCN, we believe that the R-FCN can achieve better performance when the target has been partially occluded. Several experiments are shown in Table II can verify the conception. We use Sweden Traffic Sign Dataset [25] (STSD) that consist of 2 sets in Table II experiment. Set 1 includes 1711 visible targets and 202 occluded target, and set 2 includes 1481 visible targets and 197 occluded targets. If training and testing the model only use visible ground truth, Fast-rcnn and R-FCN architecture achieve similar performance in set1 (0.9082 and 0.9038) and set2 (0.8947 and 0.8917), even Fast-rcnn slightly outperforms R-FCN. Compared with third and fourth rows, R-FCN outperforms in occluded target problem. After adding occluded targets in training and test step, R-FCN outperforms in both of set1 and set2. Table II prove that the R-FCN tend to achieve better result in target occluded case.

IV. CONCLUSION

Five detection models which are able to detect traffic sign are presented and compared in this paper. The paper can be served as a guide for selecting a traffic sign detection architecture, and the paper details the benefits and drawbacks of each meta-architecture, which can help the readers selecting a proper detection architecture for their specific tasks as well. Also, the experiment proves that 1) The behavior of detection model is strongly affected by the choice of network architecture and feature extractor 2) compared with Faster R-CNN and SSD architecture, R-FCN can obtain higher accuracy in traffic sign detection task 3) SSD perform badly in detect small-scale object as well as SSD cannot produce the precise region of the traffic sign and 4) the R-FCN can achieve better performance in target occluded case.

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